



Generation Interconnection

Combined Feasibility / System Impact Study Report

for

Queue Project AF2-045

CAMBRIA NUG 115 KV

2 MW Capacity / 20 MW Energy

July 2020

Table of Contents

1	Introduction.....	3
2	Preface.....	3
3	General	4
4	Point of Interconnection.....	4
5	Cost Summary	4
6	Transmission Owner Scope of Work	5
7	Schedule.....	5
8	Transmission Owner Analysis.....	5
8.1	Power Flow Analysis	5
9	Interconnection Customer Requirements.....	5
9.1	System Protection.....	5
9.2	Compliance Issues and Interconnection Customer Requirements	6
9.3	Power Factor Requirements.....	6
10	Revenue Metering and SCADA Requirements	7
10.1	PJM Requirements	7
10.2	Interconnected Transmission Owner Requirements.....	7
11	Summer Peak - Load Flow Analysis	8
11.1	Generation Deliverability	8
11.2	Multiple Facility Contingency	8
11.3	Contribution to Previously Identified Overloads.....	8
11.4	Potential Congestion due to Local Energy Deliverability.....	8
11.5	System Reinforcements - Summer Peak Load Flow - Primary POI.....	8
11.6	Queue Dependencies	9
11.7	Contingency Descriptions.....	9
12	Light Load Analysis	10
13	Short Circuit Analysis.....	10
14	Stability and Reactive Power	10
15	Affected Systems	10
15.1	NYISO	10
16	Attachment 1: One Line Diagram	11

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Mid-Atlantic Interstate Transmission, Inc. (MAIT, PENELEC Zone).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

3 General

The Interconnection Customer (IC), has proposed a Storage generating facility located in Cambria County, Pennsylvania. The installed facilities will have a total capability of 20 MW with 2 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 28, 2020. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-045
Project Name	CAMBRIA NUG 115 KV
State	Pennsylvania
County	Cambria
Transmission Owner	MAIT (PENELEC Zone)
MFO	20
MWE	20
MWC	2
Fuel	Storage
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

The project will reuse a portion of the capacity injection rights of the deactivated Cambria NUG and will connect to the FirstEnergy transmission using the existing generator lead line and main GSU transformer of the retired unit. As such, there are no new interconnection facilities required for this project.

Attachment 1 shows a one-line diagram of the proposed primary Direct Connection facilities for the AF2-045 generation project to connect to the FirstEnergy (“FE”) Transmission System. The IC will be responsible for constructing the facilities on its side of the Point of Interconnection (POI), including the Attachment Facilities which connect the generator to the FE Transmission System’s Direct Connection facilities.

5 Cost Summary

The AF2-045 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0
Total System Network Upgrade Costs	\$ 0
Total Costs	\$ 0

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Cost allocations for any System Upgrades will be provided in the System Impact Study Report.

6 Transmission Owner Scope of Work

There is no attachment facility, direct, or non-direct connection scope of work required.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0

7 Schedule

The project will reuse the interconnection facilities for the deactivated Cambria NUG plant and will not require any new facilities to be constructed. There is no Attachment Facilities or Direct and/or Non-Direct Connection facilities scope of work.

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF2-045 project did not contribute to any overloads on the <100 kV FE transmission system.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 115 kV circuit breaker to protect the AF2-045 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF2-045 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE Transmission System.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE Transmission System.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Interconnected Transmission Owner Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

11 Summer Peak - Load Flow Analysis

The Queue Project AF2-045 was evaluated as a 18.0 MW (Capacity 2.0 MW) uprate to the existing Cambria NUG which is an injection at the Cambria Slope 115 kV substation in the MAIT (PENELEC Zone) area. Project AF2-045 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-045 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

11.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

11.5 System Reinforcements - Summer Peak Load Flow - Primary POI

None

11.6 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

None

11.7 Contingency Descriptions

None

12 Light Load Analysis

Not required for this project.

13 Short Circuit Analysis

The following Breakers are overdutied:

None.

14 Stability and Reactive Power

Summary of the VAR requirements based upon the results of the dynamic studies:

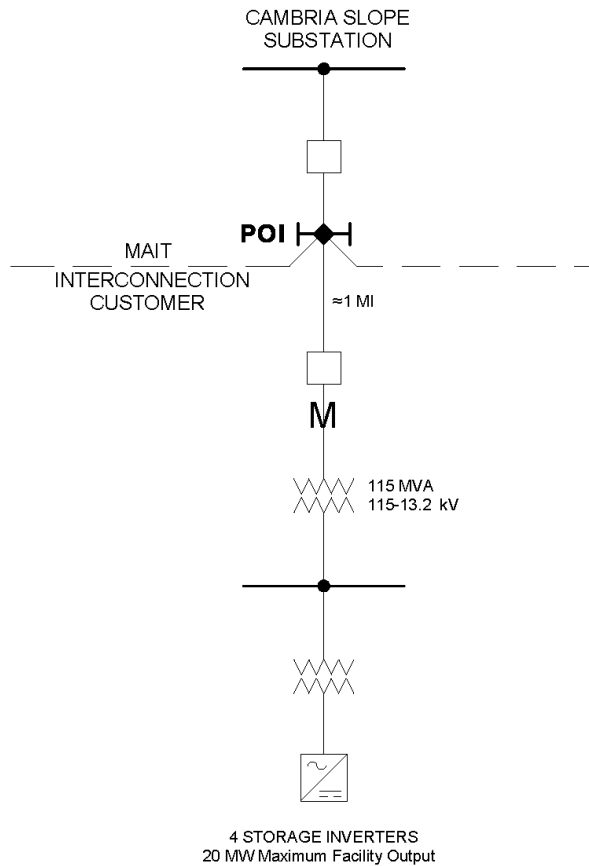
No impacts.

15 Affected Systems

15.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

16 Attachment 1: One Line Diagram



◆ = POI (POINT OF INTERCONNECTION) IS AT THE MAIT OWNED SUBSTATION DEAD-END STRUCTURE WHERE INTERCONNECTION CUSTOMER'S TRANSMISSION LINE TERMINATES

M = REVENUE METERING FOR INTERCONNECTION CUSTOMER IS OWNED, OPERATED AND MAINTAINED BY INTERCONNECTION CUSTOMER.

FirstEnergy

Energy Delivery Technical Services

TITLE
MOMENTUM ENERGY STORAGE PARTNERS, LLC INTERCONNECTION
TO THE MAIT OWNED CAMBRIA SLOPE SUBSTATION

BY: LDM DATE: 6/18/2020
APP: - ISSUE: PRELIMINARY

AGREEMENT

DOC. ID: POI-MAIT-PN-AF2-045
REV: -